Open Lake vs. Sheltered Lake: What Two Buoys Reveal

Comparing NOAA buoy records in Lake Michigan and Muskegon Lake with Python

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Buoys analyzed: 45029 (open Lake Michigan, off Holland, MI) and 45161 (GLERL buoy, inside sheltered Muskegon Lake). Data were retrieved, cleaned, analyzed and mapped in Python using pandas, Matplotlib, Cartopy and bathymetry data.

1. Buoy Locations: Aerial Imagery and Bathymetry

Two buoys were selected to give a contrast between an open-lake environment and a sheltered inland-lake environment: Buoy 45029, moored in open Lake Michigan just off the coast near Holland, MI (marked in red), and Buoy 45161, a GLERL buoy sitting inside Muskegon Lake, a drowned river-mouth lake connected to Lake Michigan by a narrow channel (marked in blue).

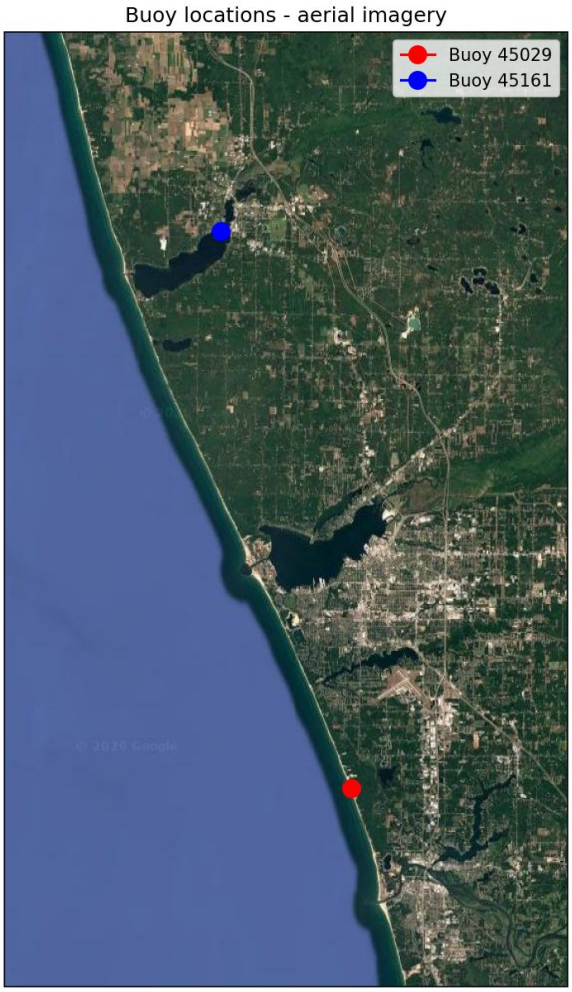


Figure 1a. Satellite/aerial imagery showing the two buoy locations along the western Michigan shoreline.

The aerial image confirms the geography described above. Buoy 45161 (blue) sits well inland, inside the enclosed body of Muskegon Lake, separated from the open water of Lake Michigan by a strip of land and a narrow connecting channel. Buoy 45029 (red) sits directly in open Lake Michigan, just offshore of a narrow sandbar/beach, with no land sheltering it from wind or wave action coming off the lake. This difference in exposure is the basis for most of the comparisons made in later sections: 45161's readings should generally reflect calmer, more sheltered conditions, while 45029's should reflect open-water conditions.

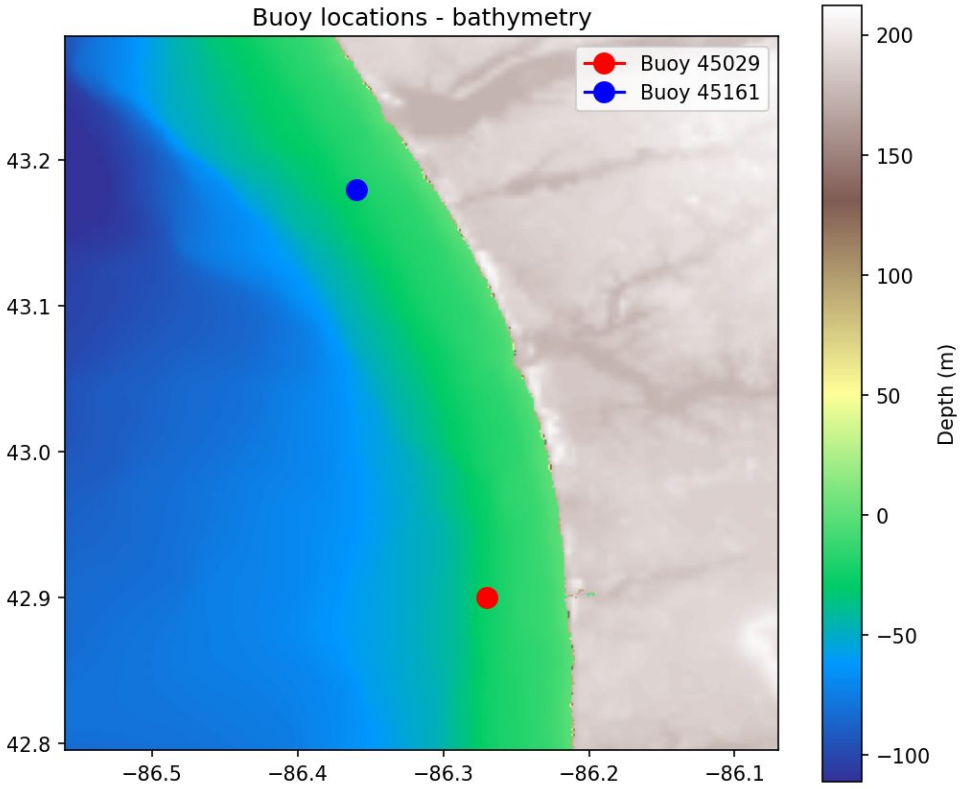


Figure 1b. Bathymetric map of the same region, with depth (m) shown as a color scale (negative = underwater, positive = land).

The bathymetry map shows the underwater shelf dropping off steadily to the west of the shoreline, from about 0 m near shore to over -100 m within roughly half a degree of longitude offshore. Buoy 45029 sits close to shore in shallower water (roughly 0 to -20 m range based on its position on the color scale), while Buoy 45161, despite being farther inland on the map, sits within the enclosed Muskegon Lake basin rather than on the open-lake shelf. The bathymetry product used here only resolves depth for the main lake body; the smaller inland lake shows as part of the same shallow near-shore green band, which is a resolution limitation of the dataset rather than a reflection of Muskegon Lake's actual depth.

2. Water Temperature Comparison

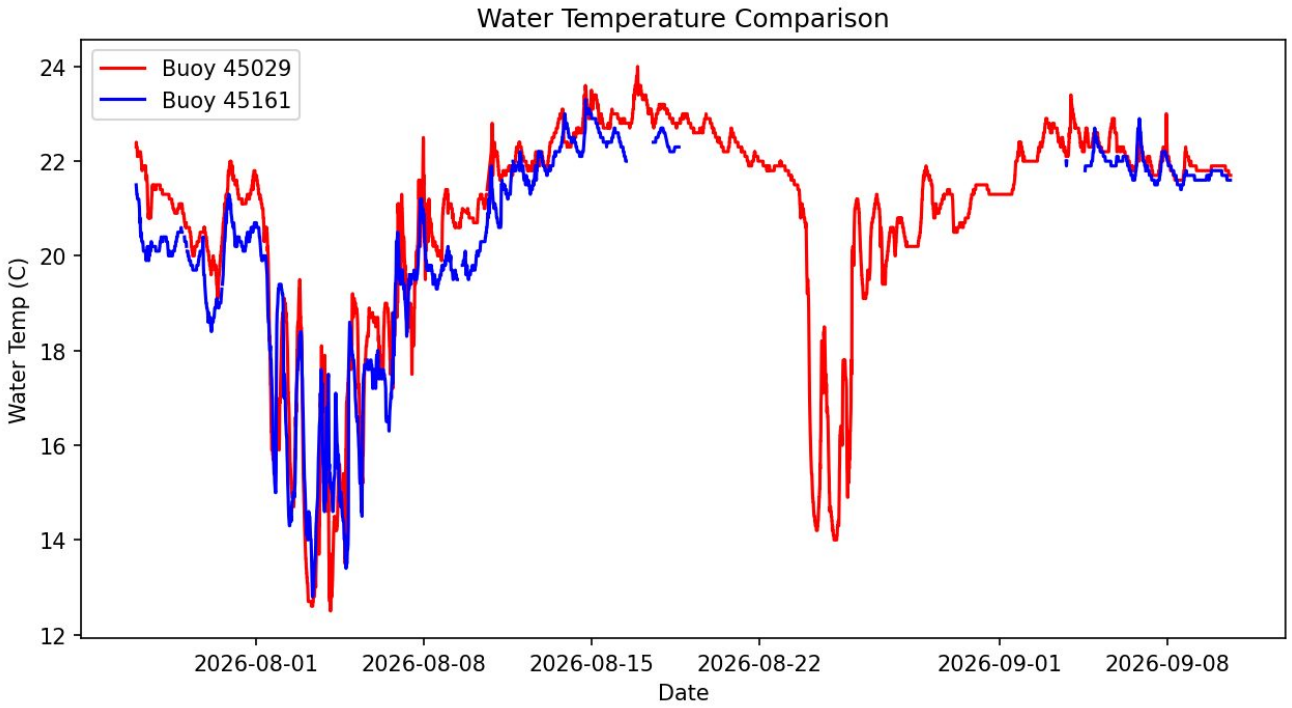


Figure 2. Water temperature (°C) over the ~45-day record for both buoys.

Both buoys track each other closely for most of the record, rising from around 20°C in late July to a peak near 24°C in mid-August, before cooling again into September. Both series also show the same sharp cold-water dips at the same times (e.g. around August 1-4 and August 23-25), which are consistent with wind-driven upwelling events pulling colder deep water to the surface near shore during storms.

The most important feature of this plot is the gap in Buoy 45161's data between roughly August 18 and September 3: the blue line is missing during this window because 45161's record has far fewer readings than 45029's (1,908 rows vs. 6,279 rows over the same period), not just a handful of missing values. The plotted line was corrected to break during this gap rather than drawing a straight diagonal line across it, since a straight line would falsely suggest a smooth, continuous reading through that period. Where data does exist for both buoys, 45161 (sheltered, inland) runs slightly cooler on average and shows marginally less extreme swings than 45029 (open lake), which fits with a smaller, more enclosed water body responding somewhat differently to wind mixing than the open lake.

3. Three-Day Rolling Average Water Temperature (Buoy 45029)

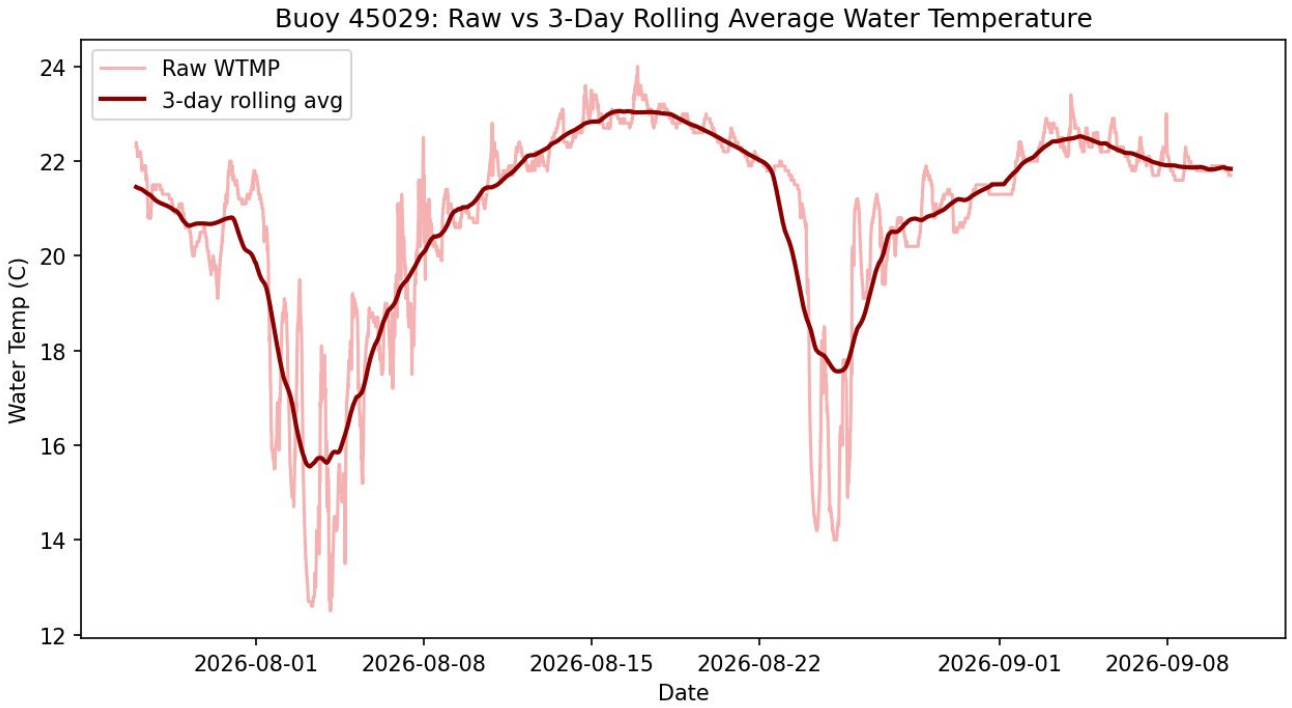


Figure 3. Raw water temperature (light red) vs. a centered 3-day rolling average (dark red) for Buoy 45029.

The raw water temperature signal (light red) is noisy on an hourly/sub-daily timescale, with frequent small oscillations layered on top of the overall trend. The 3-day rolling average (dark red, computed using a window of 432 readings at the buoy's ~10-minute sampling interval) smooths out this short-term noise and makes the underlying multi-day trend much clearer: a steady warming trend from late July into mid-August, a plateau around 23°C in mid-August, two distinct cooling dips (early August and late August, corresponding to the upwelling events also visible in Section 2), and a gradual recovery into September. The rolling average is especially useful for identifying the two cold-water events as genuine multi-day features rather than single noisy readings, since the smoothed line clearly dips well below the surrounding trend at both points.

4. Wind Speed vs. Gust Speed (Buoy 45029)

Wind speed (WSPD) and gust speed (GST) were chosen for this comparison, since gusts are, by definition, short-term spikes above the sustained wind speed and should therefore be strongly correlated with it.

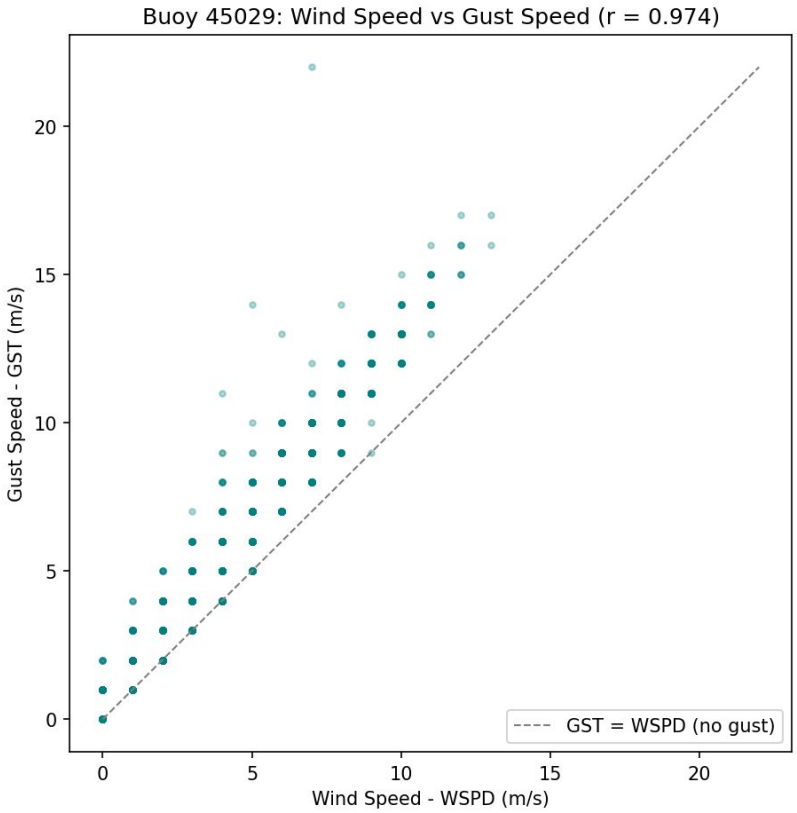


Figure 4. Scatter plot of wind speed vs. gust speed for Buoy 45029, with the 1:1 reference line (GST = WSPD) shown as a dashed line.

The Pearson correlation coefficient between WSPD and GST is r = 0.974, indicating a very strong positive linear relationship, exactly as expected given the physical link between the two variables. The points form a tight diagonal band, and nearly all of them sit on or above the dashed GST = WSPD reference line, which makes physical sense since a gust cannot be lower than the sustained wind speed at that moment. One clear outlier is visible at roughly WSPD = 7 m/s, GST = 22 m/s, well above the rest of the point cloud; this was checked against its timestamp and kept in the analysis, since a single point does not meaningfully change a correlation this strong.

5. Wind Speed vs. Wave Height (Buoy 45029)

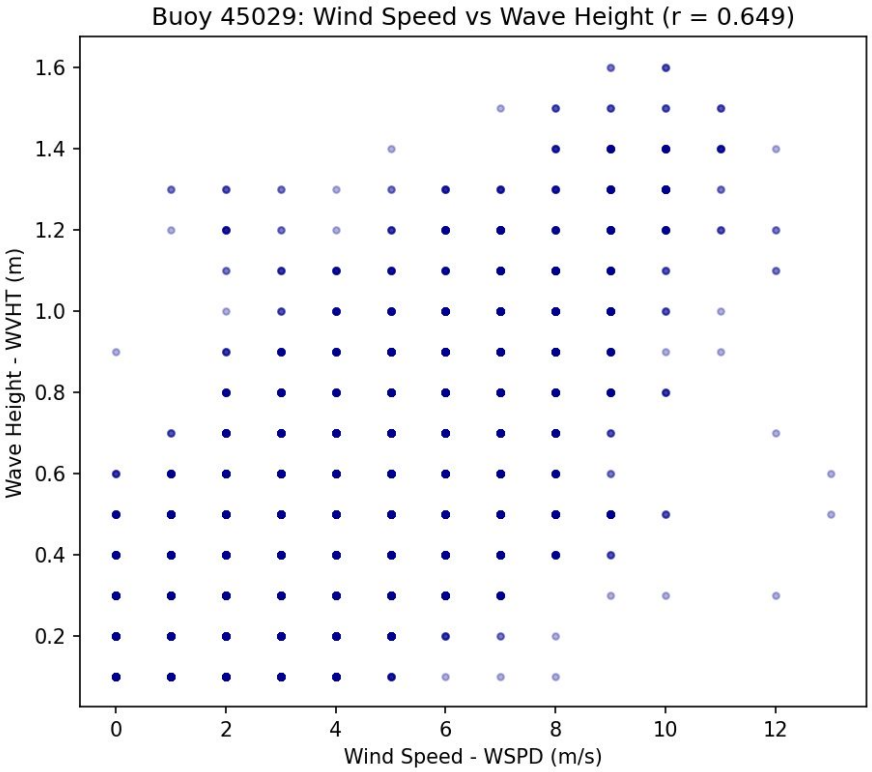


Figure 5. Scatter plot of wind speed vs. wave height for Buoy 45029.

The correlation between wind speed (WSPD) and wave height (WVHT) is r = 0.649, a moderate-to-strong positive relationship, but noticeably weaker and more scattered than the wind-vs-gust relationship in Section 4. This is physically reasonable: wave height does not respond instantaneously to wind speed the way a gust does. It also depends on fetch (the distance wind blows uninterrupted over open water), how long the wind has been blowing (duration), and water depth, so the same wind speed can correspond to a range of wave heights depending on recent conditions. The wider spread of points around the general upward trend in this scatter plot, compared to the tight band in Figure 4, visually reflects that additional dependence on fetch and duration.

6. Peak Gust Events at Both Buoys

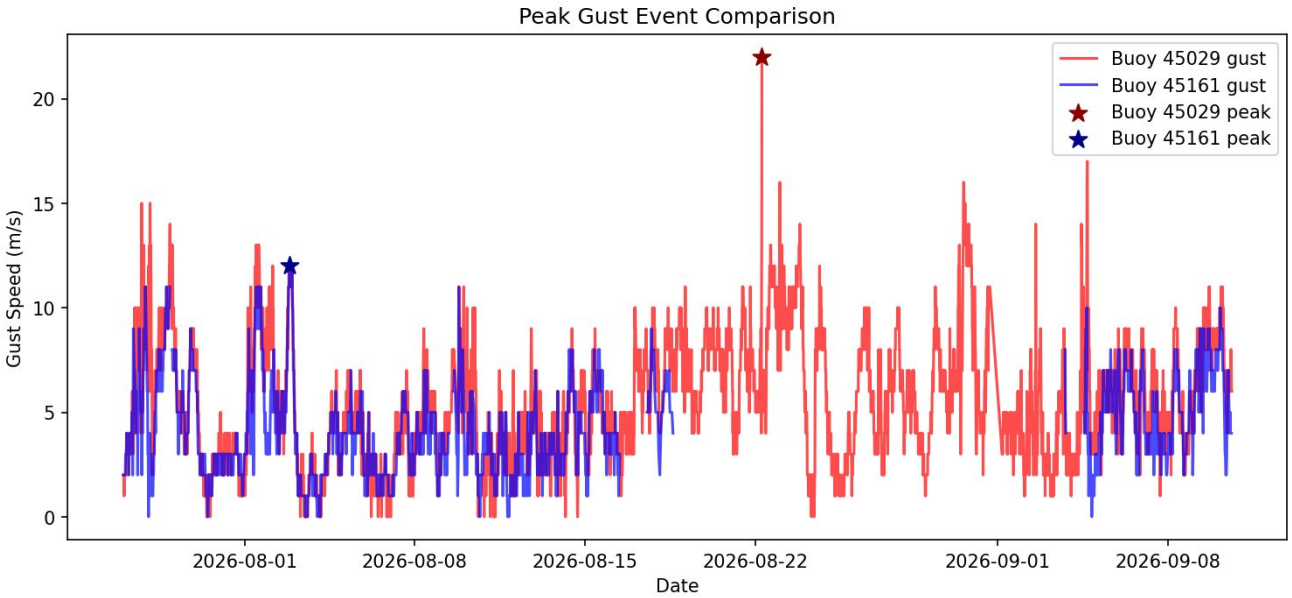


Figure 6. Gust speed over time for both buoys, with each buoy's single highest gust marked with a star.

Buoy 45029's highest recorded gust (about 22 m/s) occurred around August 20, marked with the dark red star. Buoy 45161's highest recorded gust (about 12 m/s) occurred earlier, around August 2, marked with the dark blue star. Critically, August 20 falls inside Buoy 45161's data gap (roughly August 18 to September 3, the same gap discussed in Section 2), which is visible in this plot as the long, isolated, near-flat diagonal segment of the blue line. Because 45161's sensor was not recording during that window, its peak gust of 12 m/s is only the highest value among the data it happened to have, not necessarily its true peak for the 45-day period. As a result, this comparison cannot confirm or rule out whether Buoy 45161 experienced the same storm system that produced 45029's peak gust on August 20; the apparent 18-day gap between the two peaks is most likely an artifact of missing data rather than evidence of two separate weather events. For the periods where both buoys do have overlapping data (for example, the shared gust spike visible around August 2-3), the two records track each other closely, which does support the two sites generally experiencing the same regional wind events when both sensors are actually reporting.